Wang Luo 罗旺

Ph.D. Student in Computer Science, Sun Yat-sen University, Guangzhou, China

luow69@mail2.sysu.edu.cn
robwonn.github.io
Google Scholar RobWonn

Research Interests

- **LLM reasoning** under conflicting evidence
- **Medical agents** self-improving agents for medical decision making
- **Spatial intelligence** 3D completion, world models

Education

Sun Yat-sen University — Sep 2024 – Present
Ph.D. in Computer Science (direct entry), advised by Prof. Di Wu — Guangzhou, China

Jilin University — Sep 2020 – Jun 2024
B.Eng. in Software Engineering — Changchun, China

Industry Experience

Tencent Hunyuan — Jan 2026 – Apr 2026
Research Intern — China

- Used a pretrained world model to supply occluded-region context for point cloud semantic scene completion; proposed **BelieveSSC**, which uses the semantic composition of the complete scene as the interface between the two.
- Built the paired image–point-cloud benchmark **3D-FRONT-PC-RGB**; mIoU improves by 7.7%–10.3% over the previous best method on three benchmarks. First-author paper under review at AAAI 2027.

Selected Research

Format-robust evidence reasoning in LLMs

EMNLP 2026 Oral; ICLR 2027 (under review)

- LLMs resolving knowledge conflicts change their preference when the same evidence is rendered as text, tables, or knowledge graphs. **FaDe** treats format as a nuisance variable and distills format-marginalized teacher targets into a student; it reduces format bias on ConflictBank and two controlled conflict sets across four backbones with single-pass inference.
- In medicine, built **PMC-ConForm** from PMC-Patients with source and format varied independently; source preference of LLMs shifts and can reverse under content-preserving format changes. **ForMeDER** self-distills cross-format consensus without external labels and reduces the shift on 8 LLMs, generalizing to unseen formats.

Self-improving clinical diagnosis agents

Ongoing

- Interactive diagnosis agents that take history, order tests and diagnose on MIMIC-IV emergency department cases, and revise their own diagnostic harness from their errors. Studies whether such self-improvement holds on unseen patients.

Multimodal and streaming point cloud completion

AAAI 2026; ACM MM 2026; AAAI 2027 (under review)

- **PGNet**: completion-by-correction, which starts from an image-to-3D shape prior and corrects it in feature space; average Chamfer Distance -23.5% and F-score $+7.1\%$ on ShapeNet-ViPC.
- **CRCNet**: VLM-generated part phrases form text-routed vision prototypes for point-wise refinement; best results on ShapeNet-ViPC, generalizes to unseen categories and KITTI scans.
- **OrbitScan** / **CommitStream**: the first streaming completion benchmark with sequence-level reliability metrics, and a training-free commitment operator that improves held geometry and reduces take-back on all 13 backbones evaluated.

Publications

FIRST-AUTHOR

[1] Yue Yu*, **Wang Luo***, Shenghao Ye, Zepeng Bao, Jiacheng Liu, Yongqi Li, Wenxin Tang, Tieyun Qian. Format as Nuisance: Intervention-Inspired Cross-Format Distillation for Knowledge Conflicts. **EMNLP 2026 Main**, **Oral**, **Paper Award Nomination**. code

[2] **Wang Luo**, Hongyu Chen, Di Wu, Yinlin Zhu, Hengyuan Na, Shenghao Ye, Yue Yu, Wei-Jin Huang, Miao Hu, Guocong Quan. Beyond Presentation: Toward Format-Robust Medical Evidence Reasoning in LLMs. *Under review at ICLR 2027.*

[3] **Wang Luo**, Di Wu, Hengyuan Na, Yinlin Zhu, Miao Hu, Guocong Quan. Rethinking Multimodal Point Cloud Completion: A Completion-by-Correction Perspective. **AAAI 2026**. code

[4] **Wang Luo**, Di Wu, Hengyuan Na, Yinlin Zhu, Shenghao Ye, Miao Hu, Guocong Quan. Compose, Route, Complete: Text-Routed Vision Prototypes for Multimodal Point Cloud Completion. **ACM MM 2026**.

[5] **Wang Luo**, Di Wu, Jiahao Li, Yinlin Zhu, Hengyuan Na, Shenghao Ye, Yuqi Wang. Believe, Then Complete: Reading a World Model's Belief for Semantic Scene Completion. *Under review at AAAI 2027.*

[6] **Wang Luo**, Di Wu, Yinlin Zhu, Hengyuan Na, Shenghao Ye, Wei-Jin Huang, Miao Hu, Guocong Quan. Towards Reliable Streaming Point Cloud Completion: A Benchmark and a Training-Free Commitment Operator. *Under review at AAAI 2027.*

[7] **Wang Luo**, Di Wu, Xianzhi Zhang, Yinlin Zhu, Hengyuan Na, Hongyu Chen, Shenghao Ye, Wenxin Tang, Miao Hu, Guocong Quan. A Survey of Deep Point Cloud Completion: Prior Sources, Benchmarks, and Open Challenges. *Under review at IJCV.*

* Equal contribution.

CO-AUTHORED

[1] Rubric-Guided Process Reward for Stepwise Model Routing. **EMNLP 2026 Main**.

[2] Rethinking Stepwise Model Routing: A Cost-Efficient Table Reasoning Perspective. **EMNLP 2026 Main**.

[3] DCVD: Dual-Channel Cross-Modal Fusion for Joint Vulnerability Detection and Localization. **NeurIPS 2026**.

[4] NTIRE 2025 Challenge on Short-form UGC Video Quality Assessment and Enhancement: Methods and Results. **CVPR 2025 Workshops**.

[5] Rep-Bicubic++: Enabling 8K Real-Time Super-Resolution on Edge Devices. **IEEE EdgeCom 2025**.

Further co-authored preprints on multimodal graph learning, volumetric video streaming and code understanding are listed on Google Scholar.

Honors and Awards

Paper Award Nomination (Oral), EMNLP 2026	2026
First-Class Scholarship, Sun Yat-sen University (twice)	2024, 2026
National Scholarship, Ministry of Education of China	2021–2022
First-Class Scholarship, Jilin University (four times)	2020–2024